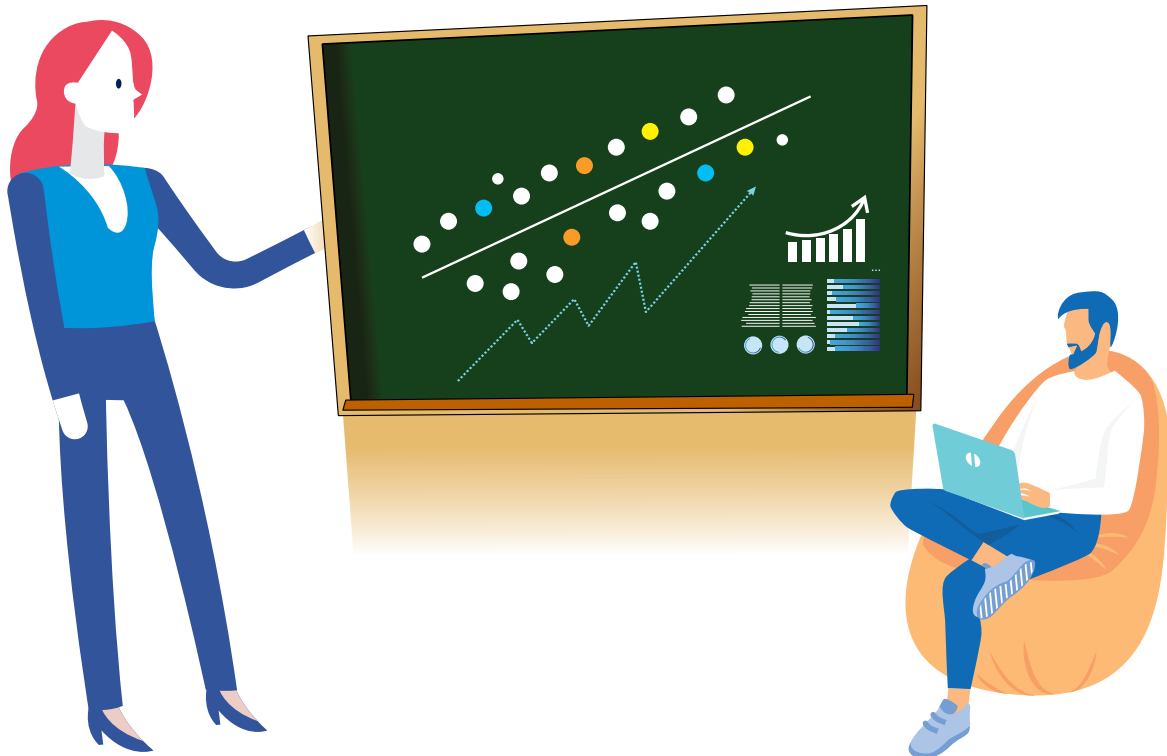


R Series: Regression

In this seventeenth article in the R series, we shall learn about linear regression in R.



In statistics, regression is a way to model the relationship between two variables. We will use R version 4.2.1 installed on Parabola GNU/Linux-libre (x86-64) for the code snippets.

```
$ R --version
```

```
R version 4.2.1 (2022-06-23) -- "Funny-Looking Kid"
Copyright (C) 2022 The R Foundation for Statistical Computing
Platform: x86_64-pc-linux-gnu (64-bit)
```

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The mathematical equation for a linear regression between two variables 'x' and 'y' is defined as 'y = ax + b', where 'a' and 'b' are constants.

The `lm()` function is used to fit linear models. Consider two vectors 'x' and 'y' that represent odd and even numbers respectively. We can fit a regression equation between them using the `lm()` function, as follows:

```
> x <- c(1, 3, 5, 7, 9)
> y <- c(2, 4, 6, 8, 10)
> r <- lm(y~x)
> print(r)
```

```
Call:
lm(formula = y ~ x)
```

```
Coefficients:
(Intercept)          x
            1            1
```

The `lm()` function accepts the following arguments: